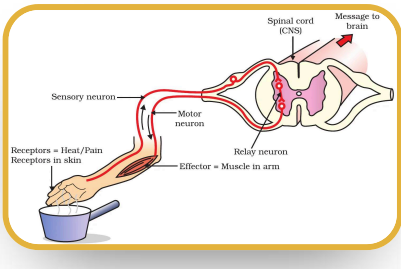
MT Classes, Motihari

Control and Coordination

Class 10 Science, Chapter 6 (NCERT, CBSE 2026-27)



Neurons, synapses, reflex arcs, the brain, plant movements, plant hormones and the endocrine glands, explained beside every diagram of the textbook, with a nerve-impulse tracer, a reflex race, a reaction-time test, a touch-me-not you can tap, an auxin light lab, an adrenaline simulator, an insulin feedback loop, real CBSE board questions with answers, every NCERT question solved and a quiz.

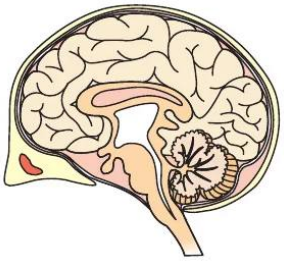
📌 Chapter snapshot

- Chapter 6 of NCERT Class 10 Science (reprint 2026-27), 13 pages, in Unit II “World of Living” (25 marks in the 80-mark theory paper, with Life Processes, How do Organisms Reproduce? and Heredity).
- Asked **every year**: a **brain part** MCQ (cerebellum, medulla), an **Assertion–Reason** on reflexes, **reflex arc** (3 marks, 2024), **sensory vs motor nerve** (2026), **touch-me-not vs tendril** (2025), **auxin** (2025), **thyroxin and goitre** (2024) and **insulin and feedback** (2023).
- Diagrams to practise: **neuron** (Fig. 6.1a, asked in NCERT Exercise 5), **reflex arc** (6.2), **human brain** (6.3). Know the positions of the **endocrine glands** (6.7).

6 Why do we need control and coordination?

We think of **movement** as a sign of life. Some movements are the result of **growth** (a seedling pushing up through soil); others are **not connected with growth** (a cat running, children on swings, a buffalo chewing cud).

- Movement is usually a **response to a change in the environment** (a **stimulus**): the cat runs because it has seen a mouse.
- Organisms also try to **use** changes to their advantage: plants grow out into sunshine; buffaloes chew cud to break tough food.
- When bright light falls on our eyes or we touch a hot object, we detect it and move to **protect ourselves**.



Chapter opener: the brain, the main coordinating centre

Every movement is **carefully controlled**: each kind of change produces the **right** response (we whisper to a friend in class instead of shouting). So living organisms need **systems providing control and coordination**, made of specialised tissues.

🐾 Animals
Nervous system (fast, electrical) + **hormones** of the endocrine system (slower, chemical). Muscles do the final action.

🌱 Plants
No nerves, no muscles. They use **electrical-chemical signals** from cell to cell and **plant hormones**; cells move by **changing their water content**.

🎯 Purpose
Detect a change, decide, and respond with **only the correct movement**, at the right time and place.

6.1 Animals: the nervous system

Control and coordination in animals is done by **nervous** and **muscular** tissue (studied in Class 9).

Receptors: where information comes in

- All information from the environment is detected by the **specialised tips of some nerve cells**, called **receptors**.
- They are usually in our **sense organs**: inner ear (hearing, balance), nose, tongue, eyes, skin.
- **Gustatory receptors** detect **taste** (tongue); **olfactory receptors** detect **smell** (nose). **Photoreceptors** (eye), **phonoreceptors** (ear) and **thermoreceptors** (skin) detect light, sound and heat.

The neuron: structure (Fig. 6.1 a)

- **Dendrites**: short, branched extensions of the cell body. Information is **acquired** at the dendritic tips.
- **Cell body** (cyton): contains the **nucleus** and cytoplasm.
- **Axon**: a long fibre that carries the impulse **away** from the cell body. It is covered by an insulating sheath.
- **Nerve ending** (axon terminal): here the electrical impulse releases **chemicals**.

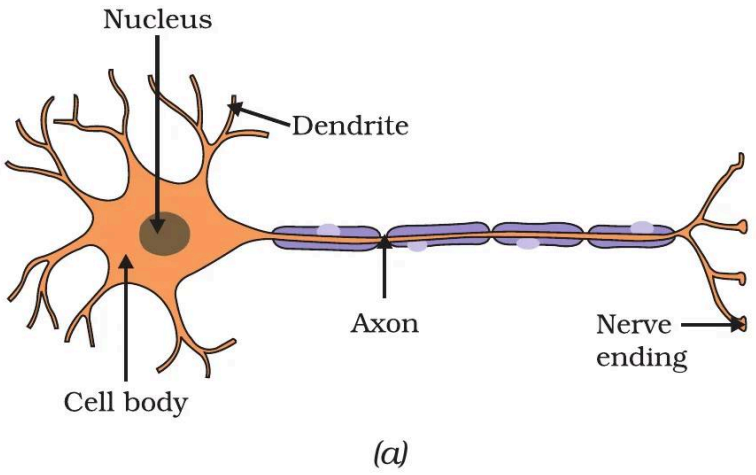


Fig. 6.1 (a) Structure of neuron

🧠 Remember the one-way path

Dendrite → **C**ell body → **A**xon → **N**erve ending → synapse. ("**D**o **C**ats **A**lways **N**ap?")

How a nerve impulse travels

1. Information acquired at the **dendritic tip** sets off a **chemical reaction** that creates an **electrical impulse**.
2. The impulse travels from the **dendrite → cell body → along the axon** to its end.
3. At the end of the axon, the impulse sets off the **release of chemicals** (neurotransmitters).
4. These chemicals **cross the gap**, called the **synapse**, and start a similar electrical impulse in a **dendrite of the next neuron**.
5. A similar synapse finally delivers impulses from neurons to **muscle cells or glands**: the **neuromuscular junction** (Fig. 6.1 b).

⭐ **NCERT in-text question: parts of a neuron**

(i) Information is **acquired** at the **dendrites** (dendritic tips). (ii) It travels as an **electrical impulse** through the **axon** (dendrite → cell body → axon). (iii) It is converted into a **chemical signal** at the **nerve ending (axon terminal)**, for passing across the synapse.

Neuromuscular junction (Fig. 6.1 b)

- The branched **axon endings** spread over a **muscle fibre**. The magnified box shows an axon ending full of tiny vesicles (bags of chemical) sitting on the folded surface of the muscle.
- Muscle fibres have many **mitochondria** (energy for contraction) and are supplied by **capillaries**.
- The chemical released here makes the **muscle contract**: this is how the nervous system finally acts.

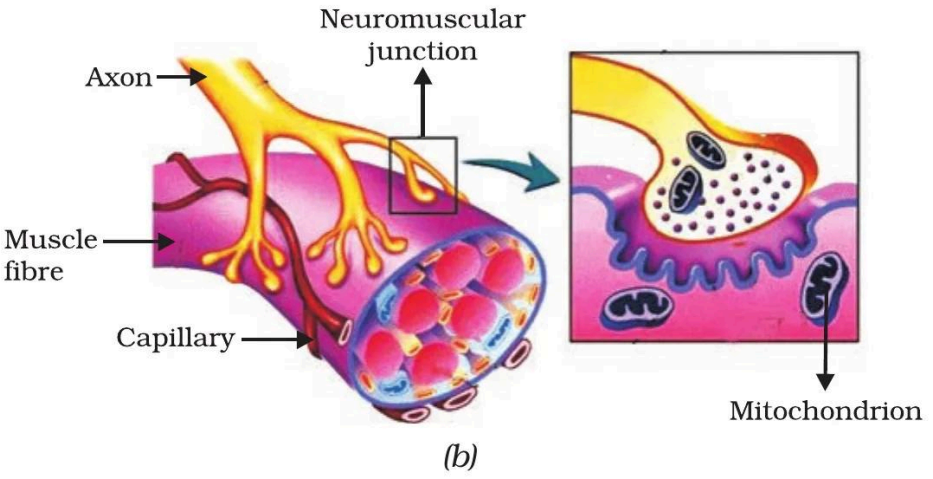


Fig. 6.1 (b) Neuromuscular junction

So **nervous tissue** is an organised network of neurons specialised for **conducting information via electrical impulses** from one part of the body to another.

Activity 6.1: taste needs smell

- Put sugar in your mouth; then block your nose and taste it again; also try eating lunch with your nose blocked.
- **Observation:** the sugar still tastes sweet, but food tastes **dull and flat**; you cannot fully enjoy its flavour. The same happens when you have a **cold**.

- **Reason:** the tongue's gustatory receptors detect only basic tastes (sweet, salty, sour, bitter, umami). The full **flavour** of food comes from smell: food vapours reach the **olfactory receptors** in the nose. With the nose blocked, the brain gets only half the information.

6.1.1 What happens in reflex actions?

We say "I pulled my hand back from the flame **reflexly**" or "my mouth watered reflexly". The common idea: we do something **without thinking** about it, or without feeling in control, yet in response to a change in the environment.

> 📖 **Definition**
>
> A **reflex action** is a **sudden, automatic and involuntary** response to a stimulus, done **without thinking**, for example pulling the hand back from a flame, blinking when something comes near the eye, the **pupil getting smaller in bright light**, sneezing, coughing.

Why not just think and then move?

- Thinking is a complex activity involving a **complicated interaction of many nerve impulses** from many neurons. The "thinking tissue" is a dense network of neurons in the **forward end of the skull**.
- Signals from the hand would have to go all the way to the brain, be thought about, and come back to the muscles. That **takes time**: we might get **burnt** first.
- Solution: connect the nerves that **detect** heat to the nerves that **move muscles** in a simpler, shorter way. This connection is a **reflex arc**.

The reflex arc (Fig. 6.2)

1. **Stimulus**: heat of the pan.
2. **Receptor**: heat/pain receptors in the skin.
3. **Sensory neuron**: carries the impulse to the spinal cord.
4. **Relay neuron** in the **spinal cord (CNS)** connects it to the motor neuron.
5. **Motor neuron**: carries the command out.
6. **Effector**: muscle in the arm contracts; the hand is pulled away (**response**).

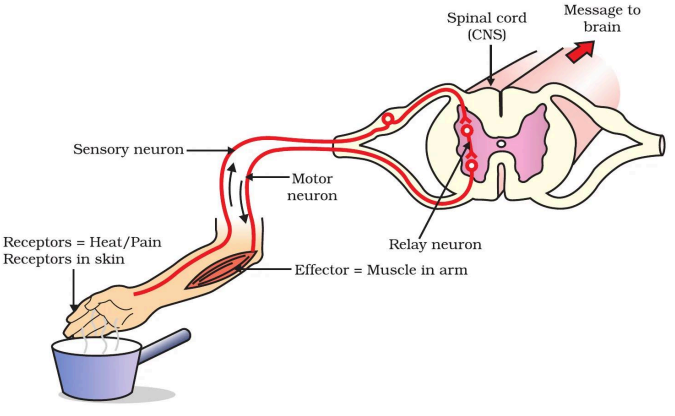


Fig. 6.2 Reflex arc

Reflex arcs are formed in the **spinal cord** itself, because nerves from all over the body meet there in a bundle on their way to the brain. The information **also goes on to the brain** (red arrow "message to brain"), so we become aware of the pain afterwards.

Stimulus → Receptor → Sensory neuron → Relay neuron (spinal cord) → Motor neuron → Effector (muscle/gland) → Response

🤔 Why did reflex arcs evolve?

The thinking process of the brain is **not fast enough**. Many animals have very little or none of the complex neuron network needed for thinking, so reflex arcs probably evolved as efficient ways of functioning **without true thought**. Even after complex brains evolved, reflex arcs stay **more efficient for quick responses**.

⭐ NCERT: trace the events when bright light is focused on your eyes

Bright light (stimulus) → **photoreceptors in the retina** detect it → impulse along the **sensory (optic) nerve** → the reflex centre in the brain (mid-brain region) → **motor nerve** → muscles of the **iris** (effector) contract → the **pupil becomes smaller**, so less light enters and the retina is protected. This happens automatically, without thinking: the **pupil reflex**.

6.1.2 The human brain

The spinal cord does more than reflexes: it is made of nerves that **supply information to think about**. Thinking needs more complex connections, which are concentrated in the **brain**, the **main coordinating centre** of the body.

Central nervous system (CNS)	Peripheral nervous system (PNS)
Brain Thinks, decides, controls voluntary and many involuntary actions	**Cranial nerves** Arise from the **brain**
Spinal cord Reflex arcs; carries messages to and from the brain	**Spinal nerves** Arise from the **spinal cord**

Job

Receives information from all parts of the body and **integrates** it

Job

Connects the CNS with all other parts of the body

Voluntary actions such as writing, talking, moving a chair or clapping are based on **deciding what to do next**. So the brain must also send messages to muscles: this is the **second way** the nervous system talks to muscles (the first is the reflex arc).

Three parts of the brain (Fig. 6.3)

Region	Parts	Main functions
Fore-brain	Cerebrum, hypothalamus	Main **thinking** part. Sensory areas for hearing, smell, sight; **association areas** that interpret information with stored memory; **motor areas** controlling **voluntary muscles**; a separate **hunger (satiety)** centre tells us we have eaten enough
Mid-brain	Mid-brain	Controls **involuntary actions** (with the hind-brain), e.g. reflex movements of the head, neck and eyes, change in pupil size

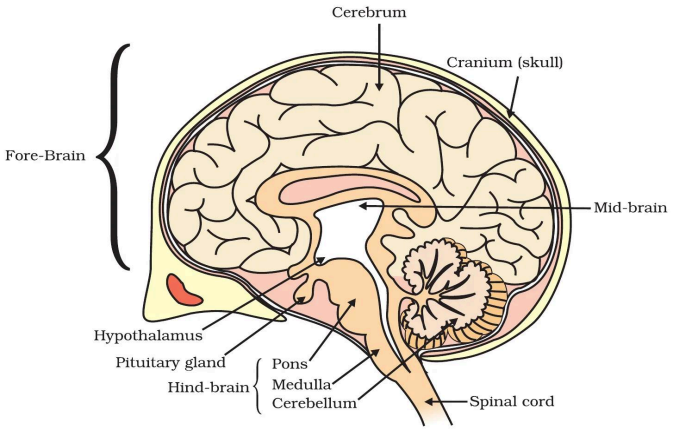


Fig. 6.3 Human brain

Region	Parts	Main functions
Hind-brain	Pons, medulla, cerebellum	**Medulla**: involuntary actions such as **blood pressure, salivation, vomiting** (also heartbeat, breathing). **Cerebellum**: **precision of voluntary actions**, **posture and balance** (walking straight, riding a bicycle, picking up a pencil). **Pons**: helps regulate breathing

Also labelled: **cranium (skull)**, **pituitary gland** (hangs below the hypothalamus), **spinal cord**.

6.1.3 How are these tissues protected?

💀 **Bony box**
The brain sits inside the **cranium (skull)**.

💧 **Fluid-filled balloon**
Inside the skull the brain is wrapped in membranes filled with **fluid** (cerebrospinal fluid) that act as a **shock absorber**.

🦴 **Backbone**
The **vertebral column** (the hard bumpy line down your back) protects the **spinal cord**.

6.1.4 How does nervous tissue cause action?

- Nervous tissue collects, sends and processes information and makes decisions, but the final movement is done by **muscle tissue**.
- Muscle cells move by **changing shape so that they shorten**.
- They contain **special proteins** (actin and myosin) that change **both their shape and their arrangement** in the cell in response to nervous electrical impulses. The new arrangement gives the muscle cell a **shorter form**: the muscle contracts.

🤔 NCERT: voluntary vs involuntary muscles, what is the difference?

Voluntary (skeletal, striated) muscles, e.g. in arms and legs, move **when we decide**: they get impulses from the **motor areas of the fore-brain**. **Involuntary** muscles (smooth muscles of the gut, blood vessels, iris; and cardiac muscle) move **on their own**, controlled by the **mid-brain and hind-brain** (and hormones); we cannot control them by thinking.

6.2 Coordination in plants

Plants have **neither a nervous system nor muscles**, yet they respond to stimuli. They show two types of movement:

Independent of growth	Dependent on growth
Example Leaves of the **sensitive plant** (chhui-mui, touch-me-not, *Mimosa*) fold and droop when touched	**Example** Root grows down, stem grows up; tendrils coil; shoots bend to light
Speed Very **quick**	**Speed** **Slow**; stops if growth is prevented
Called Immediate response (nastic movement: not in the direction of the stimulus)	**Called** Tropic movement (tropism): **directional**, towards or away from the stimulus

6.2.1 Immediate response: the sensitive plant (Fig. 6.4)

- No growth is involved, so the plant must actually **move its leaves**.
- The movement happens at a point **different from the point of touch**, so the information must be **communicated**.
- Plants use **electrical-chemical** means to pass the information **from cell to cell**, but there is **no specialised tissue** for conduction (unlike nerves).



Before touch: leaflets open



After touch: leaflets folded

- Instead of muscle proteins, plant cells **change shape by changing the amount of water** in them: they **swell or shrink**. Cells at the base of the leaflets lose water, so the leaflets fold.

Fig. 6.4 The sensitive plant

6.2.2 Movement due to growth

Tendrils (pea plant)

- Pea plants climb using **tendrils**, which are **sensitive to touch**.
- When a tendril touches a support, the part **in contact** with the object grows **less rapidly** than the part **away** from it.
- So the tendril **circles around** the object and clings to it. (This is **thigmotropism**, a growth movement towards touch.)

Tropisms: directional growth

Environmental triggers such as **light** or **gravity** change the direction in which plant parts grow. These **tropic movements** can be **towards** the stimulus (positive) or **away** from it (negative).

Tropism	Stimulus	Shoot	Root	Example
Phototropism	Light	Towards light (+)	Away from light (−)	Shoot bends to a window (Fig. 6.5)
Geotropism (gravitropism)	Gravity	Upwards, away (−)	Downwards, towards (+)	Tilted pot (Fig. 6.6)
Hydrotropism	Water	–	Towards water (+)	Roots grow towards a leaking pipe or moist soil
Chemotropism	Chemicals	Towards the chemical		**Pollen tube** grows towards the **ovule**
Thigmotropism	Touch	Towards the support		Tendrils of pea coil round a stick

Activity 6.2: response to light (Fig. 6.5)

1. Fill a conical flask with water, cover its neck with wire mesh and put 2–3 freshly **germinated bean seeds** on it.
2. Keep it in a cardboard box **open on one side**, with the open side facing light from a window.
3. After 2–3 days: **shoots bend towards light**, **roots bend away** from it.
4. Turn the flask so that the shoots face away from light. After a few days: the **old parts do not change direction**, but the **new growth** bends towards light again (shoot) and away from it (root).

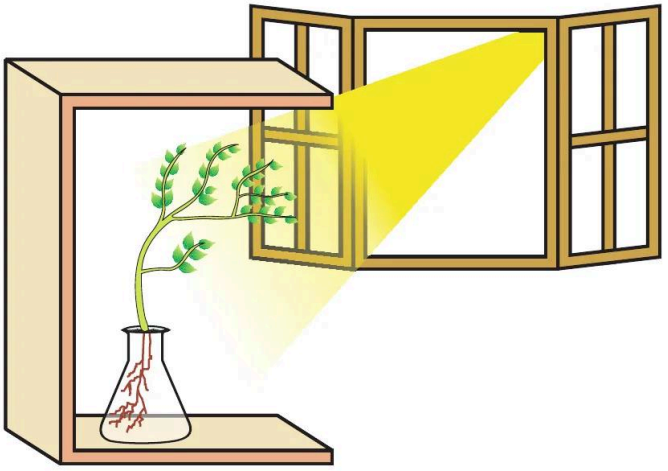


Fig. 6.5 Response of the plant to the direction of light

Conclusion: shoots are **positively phototropic** and roots **negatively phototropic**; the bending is due to **growth**, so only growing parts respond.

🤔 NCERT: how does this help the plant?

Shoots bending towards light get **more sunlight for photosynthesis**. Roots growing away from light (and down with gravity) go **deep into the soil** for **anchorage**, **water** and **minerals**.

Geotropism (Fig. 6.6)

- Roots always grow **downwards**: **positively geotropic**.
- Shoots usually grow **upwards**, away from the earth: **negatively geotropic**.
- Even when a pot is tipped on its side, the root turns down and the shoot turns up.

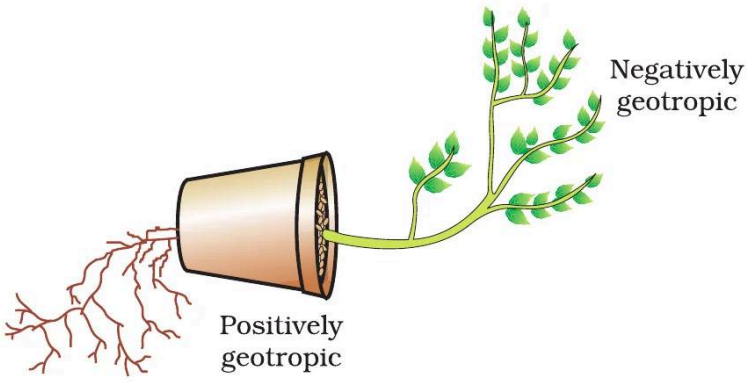


Fig. 6.6 Plant showing geotropism

Fast and slow messages

- The sensitive plant moves **very quickly**; sunflowers following day and night move **slowly**; growth movements are **even slower**.

- Animal growth is also controlled in direction: our arms and fingers grow in fixed directions, not haphazardly.
- For **fast** responses, information must travel quickly: **electrical impulses** are excellent. But they have **two limitations**:
  1. They reach **only cells connected by nervous tissue**, not every cell.
  2. After a cell fires an impulse it needs **time to reset**; cells cannot create and transmit impulses **continually**.
- So most multicellular organisms also use **chemical communication**: stimulated cells release a compound that **diffuses** all around; other cells detect it with **special molecules (receptors) on their surface**. It is **slower**, but it can reach **all cells**, regardless of nerves, and can be done **steadily and persistently**. These compounds are **hormones**.



Plant hormones (phytohormones)

Plant hormones are chemical substances that coordinate **growth, development and responses to the environment**. They are **synthesised at places away from where they act** and simply **diffuse** to the area of action.

Hormone	Type	Made / found in	Effect
Auxin	Growth promoter	Shoot tip	Helps cells **grow longer**; causes **phototropism** (bending towards light); also involved in tendril coiling
Gibberellins	Growth promoter	Various parts	Help in the **growth of the stem** (stem elongation)
Cytokinins	Growth promoter	Areas of rapid cell division: **fruits and seeds**	**Promote cell division**
Abscisic acid (ABA)	Growth **inhibitor**	Leaves, under stress	**Inhibits growth**; effects include **wilting of leaves** (a "stop growing" signal)

How auxin makes a shoot bend towards light

1. Auxin is made at the **shoot tip**.
2. When light comes from **one side**, auxin **diffuses towards the shady side** of the shoot.
3. The higher auxin concentration makes the cells on the **shady side grow longer** than those on the lit side.

4 So the shoot **bends towards light**.

> ⭐ **NCERT: design an experiment to demonstrate hydrotropism**
>
> Take a glass trough (or a wide box) filled with moist sand or soil. Plant a few germinating seeds (gram or bean) in the middle. Bury a **porous clay pot** (or a small cup with holes) filled with water on **one side** only, so that only that side stays wet. Keep the other side dry. After about a week, gently remove the sand: the **roots have bent and grown towards the water pot**. Roots are **positively hydrotropic**. (Keep the set-up in uniform light so that light does not affect the result.)

6.3 Hormones in animals

Fight or flight: adrenaline

- A squirrel in a scary situation must prepare to either **fight or run away**. Both need lots of energy and many tissues working together.
- Electrical impulses alone would reach only a **limited range of tissues**. A **chemical signal** reaches **all cells** and gives the wide-ranging changes needed.
- This is done by **adrenaline**, secreted by the **adrenal glands** (on top of the kidneys).
- Adrenaline is secreted **directly into the blood** and carried to its **target organs**:
  - **Heart beats faster** → more oxygen supplied to muscles.
  - Muscles around small arteries of the **digestive system and skin contract** → blood supply there is **reduced** and **diverted to skeletal muscles**.
  - **Breathing rate increases** because of contractions of the **diaphragm and rib muscles**.
- Such hormones are part of the **endocrine system**, the **second way** of control and coordination in our body.

Endocrine glands (Fig. 6.7, Activity 6.3)

Endocrine glands are **ductless** glands that release hormones **directly into the blood**, which carries them to target organs.

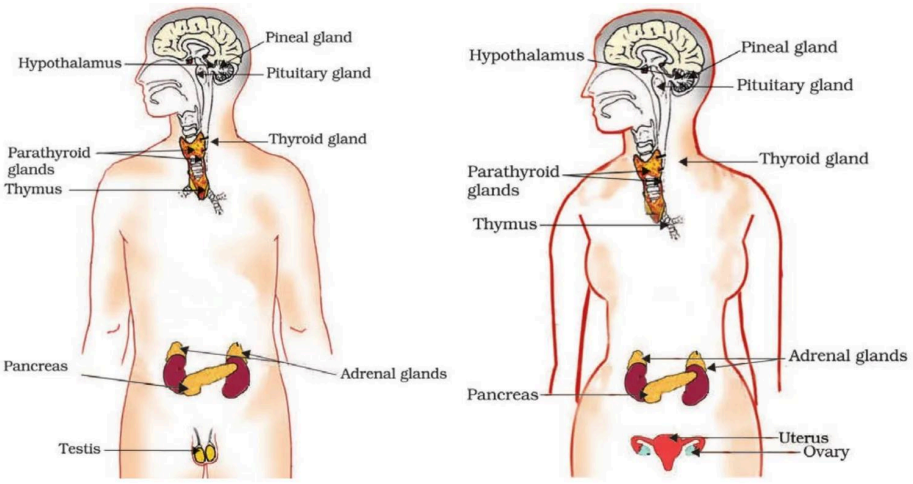


Fig. 6.7 Endocrine glands in human beings (a) male, (b) female

Hormones for coordinated growth

🧂 Thyroxin (thyroid)

Regulates **carbohydrate, protein and fat metabolism** for the best balance for growth. **Iodine** is essential to make it. Iodine deficiency → **goitre** (swollen neck, since the thyroid is in the neck). Hence **iodised salt**.

📏 Growth hormone (pituitary)

Regulates **growth and development**. Deficiency in childhood → **dwarfism**; excess → **gigantism** (very tall).

🧑 Testosterone / oestrogen

Cause the changes of **puberty** (age 10–12 onwards): testosterone in males (testes), oestrogen in females (ovaries).

🍬 Insulin (pancreas)

Regulates **blood sugar level**. Too little insulin → sugar level in blood rises → **diabetes**, treated with insulin injections.

⚡ Adrenaline (adrenal)

Prepares the body for **emergencies**: fight or flight.

🎛️ Releasing hormones (hypothalamus)

Stimulate the **pituitary** to release its hormones, e.g. **growth hormone releasing factor** when growth hormone is low.

💡 Do you know?

The **hypothalamus** plays an important role in the release of many hormones. When the level of growth hormone is low, the hypothalamus releases **growth hormone releasing factor**, which stimulates the **pituitary gland** to release growth hormone. The pituitary is often called the **master gland** because its hormones control other glands.

Feedback mechanism

Hormones must be secreted in **precise quantities**. The **timing and amount** of hormone released are regulated by **feedback mechanisms**.

> Blood sugar rises → detected by cells of the pancreas → more insulin secreted → sugar level falls → insulin secretion is reduced

Activity 6.4: Table 6.1, completed

S.No.	Hormone	Endocrine gland	Functions
1	Growth hormone	Pituitary gland	Stimulates growth in all organs
2	**Thyroxin**	Thyroid gland	Regulates metabolism for body growth
3	Insulin	**Pancreas**	Regulates blood sugar level
4	Testosterone	Testes	**Development of male sex organs and male features at puberty** (deep voice, beard)
5	**Oestrogen**	Ovaries	Development of female sex organs, regulates menstrual cycle, etc.
6	Adrenaline	Adrenal gland	**Prepares the body for emergencies: faster heartbeat and breathing, more blood to skeletal muscles**
7	Releasing hormones	**Hypothalamus**	Stimulates pituitary gland to release hormones

Other glands in Fig. 6.7 (for Activity 6.3): **pineal** (melatonin, sleep–wake cycle), **parathyroid** (parathormone, calcium level in blood), **thymus** (thymosin, development of immunity in children).



Differences to learn (2- and 3-mark favourites)

Reflex action vs voluntary action (e.g. walking) (NCERT Q)

Reflex action	Voluntary action (walking)
Automatic, done without thinking	Done consciously, after thinking and deciding
Controlled by the **spinal cord** (some by the brain stem)	Controlled by the **fore-brain (cerebrum)**; precision by the cerebellum

Reflex action	Voluntary action (walking)
Very fast	Slower
Cannot be controlled at will	Under our control: we can start, stop or change it
e.g. withdrawing hand from a flame	e.g. walking, writing

Involuntary action vs reflex action (NCERT Ex. 10)

Involuntary action	Reflex action
Goes on continuously or slowly, without our thinking, as part of normal body functions	A sudden response to a **specific stimulus**, usually to protect us
Controlled by the **mid-brain and hind-brain** (medulla)	Mostly controlled by the **spinal cord**
Not necessarily triggered by an outside stimulus	Always triggered by a stimulus
e.g. heartbeat, breathing, salivation, peristalsis	e.g. knee jerk, hand pulled from a hot object, sneezing

Nervous vs hormonal control (NCERT Ex. 11)

Nervous mechanism	Hormonal (endocrine) mechanism
Messages are **electrical impulses** (and chemicals at synapses)	Messages are **chemicals (hormones)**
Travel along **nerve fibres**	Travel through the **blood**
Very fast	**Slow**
Effect is **short-lived**	Effect is usually **long-lasting**, steady and persistent
Reaches only cells **connected by nerves**	Can reach **all cells** of the body; acts on target organs
Cells need time to reset; cannot fire continually	Can be released steadily

Similarity: both coordinate body activities and respond to stimuli; both work together (e.g. fear: nerves + adrenaline).

Sensory nerve vs motor nerve (CBSE 2026)

Sensory (afferent)	Motor (efferent)
Carries impulses **from receptors (sense organs) to the CNS**	Carries impulses **from the CNS to effectors** (muscles or glands)
Brings information in	Takes commands out

Dendrite vs axon

Dendrite	Axon
Short and highly branched; many per neuron	Long, single fibre per neuron, branched only at the end
Receives information; carries impulse **towards** the cell body	Carries impulse **away** from the cell body

Movement of the sensitive plant vs shoot towards light (NCERT Q)

Sensitive plant leaves	Shoot bending towards light
Stimulus: **touch**	Stimulus: **light**
Not due to growth	Due to **growth** (unequal growth caused by auxin)
Not directional (does not depend on direction of stimulus): nastic	**Directional**: towards light (tropic)
Very fast; reversible	Slow; permanent
Cells change shape by losing water	Cells on the shady side elongate

Sensitive plant leaf vs movement in our legs (NCERT Ex. 12)

Sensitive plant	Our legs
Involuntary response to touch	Voluntary action (we decide), except reflexes
No nerves: information passes cell to cell by electrical-chemical means	Information passes through **nerves** from the brain
Cells change shape by **changing water content** (swelling, shrinking)	Muscle cells shorten using **special proteins** that change shape and arrangement

Sensitive plant	Our legs
No muscles involved	Muscles contract and pull bones

Phototropism vs geotropism

Phototropism	Geotropism
Response to **light**	Response to **gravity**
Shoot +ve, root −ve	Root +ve, shoot −ve

Plant hormones vs animal hormones

Plant hormones	Animal hormones
No special glands; made in growing regions (tips, fruits, seeds)	Made in special **endocrine glands**
Move by **diffusion** (and through tissues)	Carried by **blood**
Control growth, direction of growth, wilting	Control metabolism, growth, puberty, sugar level, emergencies

Cerebrum vs cerebellum

Cerebrum	Cerebellum
Part of the **fore-brain**; largest part	Part of the **hind-brain**, at the back
Thinking, memory, intelligence, senses, voluntary actions	Precision of voluntary actions, posture and balance

Endocrine vs exocrine glands (extra)

Endocrine	Exocrine
Ductless; secretions go directly into blood	Have ducts that carry secretions
Secrete hormones (e.g. thyroid, pituitary)	Secrete enzymes, sweat, saliva (e.g. salivary glands)

The **pancreas** is both: insulin goes into blood (endocrine), pancreatic juice goes by a duct to the intestine (exocrine).



Practise the diagrams

Tap any diagram to open it full size. Draw each one with all its labels; board papers ask “draw a neuron” or “draw a reflex arc” for 2 to 5 marks.

✏️ Drawing tips that save marks

- **Neuron**: cell body with nucleus, many short dendrites, one long axon with sheath, branched nerve ending. Add an arrow showing the direction of the impulse (dendrite → axon).
- **Reflex arc**: receptor (skin) → sensory neuron → spinal cord with relay neuron → motor neuron → effector (muscle). Use arrows on the neurons. A cross-section of the spinal cord as a butterfly-shaped grey area is enough.
- **Brain**: label cerebrum, cerebellum, medulla, pons, mid-brain, hypothalamus, pituitary, spinal cord, cranium. Show fore-, mid- and hind-brain with brackets.
- Use a pencil, ruler-straight label lines on one side, and write the figure title below.

CBSE Previous year board questions

These questions come from the CBSE Class 10 Science board papers of the years shown, with the paper code (set) and marks. Tap a question to see a model answer.

Where we could not check a question word for word (for example its options), it is **reworded** and marked so. The years, sets and marks shown are only those we could confirm from the paper or its official marking scheme.

– 2026 · 31/1/1 · 1 mark

Human brain has various parts or regions that help in different actions, responses and coordination. From the following, identify the part responsible for precision of voluntary actions: (A) Cerebrum (B) Cerebellum (C) Medulla (D) Pons

> **(B) Cerebellum.** It is in the hind-brain and is responsible for precision of voluntary actions and for posture and balance. The cerebrum does the thinking; the medulla controls involuntary actions.

– 2026 · 31/1/1 · 1 mark

Assertion (A): Reflex actions do not involve thinking. Reason (R): Most reflex actions are controlled by the spinal cord.

> **(A) Both A and R are true, and R is the correct explanation of A.** The reflex arc connects the sensory and motor neurons in the spinal cord, so the response happens before the thinking part of the brain is involved.

– 2026 · 31/1/1 · 2 marks

State two differences between the act of chewing food and salivation on sight of food.

(1) **Chewing** is a **voluntary** action, controlled by the **fore-brain** (cerebrum); **salivation** on seeing food is **involuntary**, controlled by the **medulla** of the hind-brain. (2) Chewing needs **conscious effort** (we can stop it); salivation happens **automatically** and we cannot control it by thinking.

– 2026 · 31/1/1 · 1 mark

Differentiate between the following pair of terms: motor nerve and sensory nerve.

A **sensory nerve** carries impulses **from the receptors (sense organs) to the central nervous system** (brain or spinal cord). A **motor nerve** carries impulses **from the central nervous system to the effectors** (muscles or glands).

– 2026 · 31/1/1 · 1 mark **State the difference between nephron and neuron.**

A **nephron** is the structural and functional unit of the **kidney** that filters nitrogenous wastes from blood; a **neuron** is the structural and functional unit of the **nervous system** that carries information as nerve impulses.

– 2025 · 31/1/1 · 1 mark

The plant hormone whose concentration stimulates the cells to grow longer on the side of the shoot which is away from light is: (A) Cytokinins (B) Gibberellins (C) Adrenaline (D) Auxins

(D) Auxins. Auxin made at the shoot tip diffuses to the shady side, where cells grow longer, so the shoot bends towards light. (Adrenaline is an animal hormone.)

– 2025 · 31/1/1 · 3 marks (reworded)

Plants have neither a nervous system nor muscles, yet they respond to stimuli: the leaves of the touch-me-not plant fold up and droop when touched. (a) How is the information communicated in this plant? (b) What enables the plant cells to bring about the observable response? (c) Differentiate this movement from the movement of tendrils of a pea plant.

(a) By **electrical-chemical means**, passed **from cell to cell**; there is no specialised tissue (no nerves) for conducting information. (b) The cells **change shape by changing the amount of water** in them: they swell or shrink, so the leaflets fold. (c) Touch-me-not: movement is **not due to growth**, is very fast and **not directional**. Tendril: movement is **due to growth** (the side away from the support grows faster), slow, and **directional** (towards the support: thigmotropism).

– 2024 · 31/1/1 · 1 mark (reworded)

The point where the nerve ending of a motor neuron meets a muscle fibre, so that impulses are delivered to the muscle, is called the: (a) dendrite (b) synapse between two neurons (c) neuromuscular junction (d) cell body

(c) Neuromuscular junction (Fig. 6.1 b). (The marking scheme confirms the answer; the options here are reconstructed.)

– 2024 · 31/1/1 · 2 marks (reworded)

(A) Give two differences between the movement of the leaves of a sensitive plant and the downward movement of roots. OR (B) Name the hormone made by the thyroid gland. Why is iodised salt advised in our diet, and what disease does iodine deficiency cause?

(A) **Sensitive plant:** stimulus is **touch**; **no growth** is involved; movement is **non-directional**. **Roots growing down:** stimulus is **gravity**; **growth** is involved; movement is **directional** (geotropism). **OR** (B) **Thyroxin**, made by the **thyroid gland**. Iodine is necessary for the thyroid gland to make thyroxin, so iodised salt is advised. Deficiency of iodine causes **goitre** (swollen neck).

– 2024 · 31/1/1 · 3 marks (reworded)

What is a reflex action? Give an example. Write the sequence of events (pathway) of this reflex action as a flow chart.

Reflex action: a sudden, spontaneous, automatic response to a stimulus, without thinking, e.g. **sneezing** (or withdrawing the hand from a hot object). **Pathway:** Stimulus (dust/irritant) → **Receptors** (in the nose) → **Sensory neuron** → **Spinal cord / relay neuron** (for sneezing the reflex centre is in the medulla) → **Motor neuron** → **Effector** (muscles of chest and diaphragm) → **Response** (sneeze). The marking scheme gives this sequence: stimulus, receptors, sensory neuron, spinal cord (relay neuron), motor neuron, effector (muscles), response.

– 2023 · 31/1/1 · 2 marks (reworded)

(a) State the function of insulin and name the disease caused by its deficiency. (b) How is the timing and amount of insulin secreted regulated?

(a) Insulin, made by the pancreas, **lowers (regulates) the blood sugar level**. Its deficiency causes **diabetes**. (b) By a **feedback mechanism**: when the sugar level in blood **rises**, it is detected by cells of the pancreas, which produce **more insulin**; as the blood sugar level **falls**, insulin secretion is **reduced**.

Practice

Board-pattern practice

Extra questions in the current CBSE pattern (not from a specific year). Tap to check.

MCQs (1 mark each)

M1 **Which part of the brain controls blood pressure, salivation and vomiting? (a) Cerebrum (b) Cerebellum (c) Medulla (d) Hypothalamus**

(c) Medulla.

M2 **The hormone that promotes cell division in fruits and seeds is (a) auxin (b) cytokinin (c) abscisic acid (d) gibberellin**

(b) Cytokinin.

M3 **Growth of a pollen tube towards the ovule is an example of (a) hydrotropism (b) chemotropism (c) geotropism (d) phototropism**

(b) Chemotropism.

M4 **Which gland is called the master gland? (a) Thyroid (b) Adrenal (c) Pituitary (d) Pancreas**

(c) Pituitary. Its hormones control many other glands.

M5 **Which of these is NOT a function of adrenaline? (a) faster heartbeat (b) faster breathing (c) more blood to the digestive system (d) more blood to skeletal muscles**

(c). Adrenaline **reduces** blood flow to the digestive system and skin.

Assertion–Reason (1 mark each)

(a) Both A and R are true, and R explains A. (b) Both true, but R does not explain A. (c) A true, R false. (d) A false, R true.

1 **A: The shoot of a plant kept near a window bends towards light. R: Auxin collects on the side of the shoot facing the light.**

(c) Auxin moves to the **shady** side.

2 **A: Iodine deficiency can cause goitre. R: Iodine is needed by the thyroid to make thyroxin.**

(a)

3 **A: Hormonal signals are slower than nerve impulses. R: Hormones travel through the blood and reach all cells.**

(a) (Travel by blood and diffusion is slower than electrical conduction.)

4 **A: Movement of the leaves of the sensitive plant is a growth movement. R: The leaves droop when touched.**

(d) It is **not** a growth movement; R is true.

Short answer

S1 **A person met with an accident and the cerebellum was damaged. What problems would he face?**

He would lose **balance and posture** and could not make **precise voluntary movements**: walking in a straight line, riding a bicycle or picking up a pencil would be difficult.

S2 **Why does our heart beat faster when we are frightened?**

Adrenaline is secreted from the adrenal glands into the blood; it acts on the heart so that it beats faster, supplying more oxygen to the muscles to prepare for fight or flight.

S3 **Name the two limitations of electrical impulses that make hormones necessary.**

(1) They reach only cells **connected by nervous tissue**. (2) Cells need time to **reset**, so they cannot generate and transmit impulses **continually**.

S4 **A child's growth is much less than normal for his age. Which hormone may be deficient, and which gland makes it? What is the condition called?**

Growth hormone from the **pituitary gland**; the condition is **dwarfism**.

S5 **What is a synapse? Why does a nerve impulse travel in only one direction?**

A synapse is the **gap between the nerve ending of one neuron and the dendrite of the next**. The chemical is released only from the **axon end** and detected only by the **dendrite** of the next neuron, so the impulse passes one way only.

Case-based (4 marks)

C1 **Ravi accidentally touched a hot iron and pulled his hand away at once; only after that did he feel the pain. (i) Name this type of action. (ii) Where is the connection between the sensory and motor neuron made? (iii) Why did he feel the pain later? (iv) Name the receptor and effector involved.**

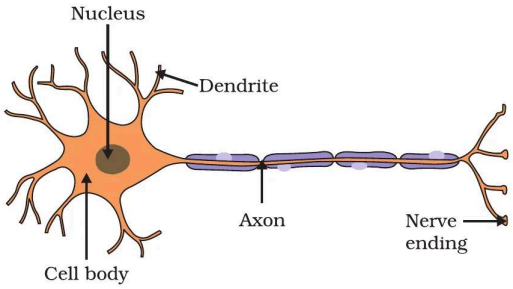
(i) **Reflex action**. (ii) In the **spinal cord**, through a **relay neuron**. (iii) The message also goes up the spinal cord to the **brain**, which makes us aware of the pain; this takes longer than the reflex. (iv) Receptor: **heat/pain receptors in the skin**; effector: **muscles of the arm**.

C2 **A student placed a potted seedling in a box with a hole on one side facing a window. After a week the shoot had bent towards the hole. (i) Name this response. (ii) Which hormone is responsible and where is it made? (iii) Explain how it causes bending. (iv) What will happen if the tip of the shoot is cut off before the experiment?**

(i) **Positive phototropism**. (ii) **Auxin**, made at the **shoot tip**. (iii) Auxin diffuses to the **shady side**; cells there grow longer than on the lit side, so the shoot bends towards light. (iv) No auxin is made, so the shoot will **not bend** (and will grow very little).

Long answer (5 marks)

– **L1** **Draw a labelled diagram of a neuron. Explain how information travels from one neuron to the next and to a muscle.**

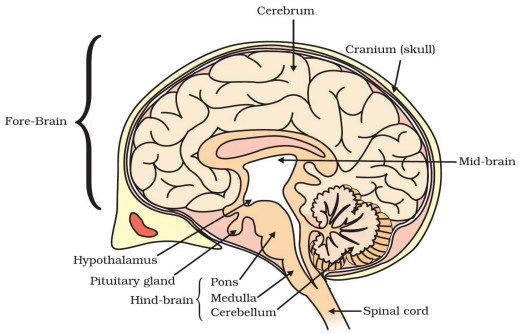


(a)

Model diagram: Fig. 6.1 (a)

Information is acquired at the **dendritic tips**, setting off a chemical reaction that creates an **electrical impulse**. It travels from the dendrite to the **cell body** and along the **axon** to its end, where it releases **chemicals**. These cross the **synapse** and start a similar impulse in the dendrite of the next neuron. At the end, a similar junction (**neuromuscular junction**) delivers the impulse to muscle cells, which contract.

– **L2** **Draw the human brain and label its parts. Write one function each of the cerebrum, cerebellum, medulla, hypothalamus and pituitary gland.**



Model diagram: Fig. 6.3

Cerebrum: thinking, memory, interpreting senses, voluntary actions. **Cerebellum**: precision of voluntary actions, posture and balance. **Medulla**: involuntary actions such as blood pressure, salivation, vomiting. **Hypothalamus**: hunger centre; releases releasing hormones that control the pituitary. **Pituitary**: secretes growth hormone (and controls other glands).

– **L3** **What are hormones? Name the gland, hormone and one function for: (a) regulating blood sugar (b) metabolism (c) emergencies (d) growth (e) male features at puberty. How is hormone secretion controlled?**

Hormones are chemical messengers secreted by endocrine glands directly into the blood, acting on target organs. (a) Pancreas: insulin, lowers blood sugar. (b) Thyroid: thyroxin, regulates carbohydrate, protein and fat metabolism. (c) Adrenal: adrenaline, faster heartbeat and breathing. (d) Pituitary: growth hormone, growth of all organs. (e) Testes: testosterone, male sex organs and features. Secretion is controlled by **feedback mechanisms**, e.g. rising blood sugar → more insulin; falling sugar → less insulin.

Recap What you have learnt

- Control and coordination are the functions of the **nervous system** and **hormones** in our bodies.
- The responses of the nervous system can be classified as **reflex action, voluntary action or involuntary action**.
- The nervous system uses **electrical impulses** to transmit messages.
- The nervous system gets information from our **sense organs** and acts through our **muscles**.
- **Chemical coordination** is seen in both plants and animals.
- Hormones produced in one part of an organism **move to another part** to achieve the desired effect.
- A **feedback mechanism** regulates the action of the hormones.

Quiz Test yourself

Tap an answer. **Score: 0 / 20**

1. Which of the following is a plant hormone? (NCERT Ex. 1)

Insulin | Thyroxin | Oestrogen | **Cytokinin**

The other three are animal hormones.

2. The gap between two neurons is called a (NCERT Ex. 2)

- Dendrite
- **Synapse**
- Axon
- Impulse

Chemicals cross the synapse.

3. The brain is responsible for (NCERT Ex. 3)

- Thinking
- Regulating the heart beat
- Balancing the body
- **All of the above**

Cerebrum, medulla and cerebellum respectively.

4. Information is acquired by which part of a neuron?

- **Dendrite**
- Axon
- Nerve ending
- Synapse

At the dendritic tip.

5. Olfactory receptors detect

- Taste
- **Smell**
- Sound
- Light

Gustatory receptors detect taste.

6. Reflex arcs are formed in the

- Cerebrum
- **Spinal cord**
- Cerebellum
- Muscles

Where input and output nerves first meet.

7. Precision of voluntary actions and body balance is maintained by the

- Medulla
- Pons
- **Cerebellum**
- Hypothalamus

CBSE 2026 MCQ.

8. Salivation, vomiting and blood pressure are controlled by the

- **Medulla**
- Cerebrum
- Mid-brain
- Spinal cord

Medulla in the hind-brain.

9. The feeling of being full after eating comes from a centre in the

- **Fore-brain**
- Mid-brain
- Hind-brain
- Spinal cord

The hunger centre (hypothalamus).

10. Cranial nerves arise from the

- Spinal cord
- **Brain**
- Heart
- Muscles

Spinal nerves arise from the spinal cord.

11. Leaves of the sensitive plant fold because cells

- Grow faster on one side
- **Lose water and change shape**
- Contract with muscle proteins
- Make auxin

No growth and no muscles involved.

12. Roots growing downwards show

- Negative geotropism
- **Positive geotropism**
- Positive phototropism
- Chemotropism

Towards gravity.

13. Auxin is synthesised at the

- Root tip
- **Shoot tip**
- Leaf base
- Seed coat

It then diffuses to the shady side.

14. A hormone that inhibits growth and causes wilting of leaves is

- Gibberellin
- Cytokinin
- **Abscisic acid**
- Auxin

A growth inhibitor.

15. Adrenaline is secreted by the

- Thyroid
- Pituitary
- **Adrenal glands**
- Pancreas

It goes directly into the blood.

16. Iodine is needed to make

- Insulin
- **Thyroxin**
- Adrenaline
- Growth hormone

Deficiency causes goitre.

17. Deficiency of growth hormone in childhood causes

- Goitre
- Diabetes
- **Dwarfism**
- Gigantism

Growth hormone comes from the pituitary.

18. Insulin is produced by the

Liver	**Pancreas**
Kidney	Adrenal gland

It regulates blood sugar.

19. Growth hormone releasing factor is released by the

Hypothalamus	Pituitary
Thyroid	Pineal

It stimulates the pituitary.

20. The timing and amount of hormone released are regulated by

Reflex arcs	**Feedback mechanisms**
Synapses	Tropisms

e.g. blood sugar and insulin.

NCERT Answers to all questions

Every question from the chapter, with answers you can write in the exam. Tap to open.

Questions (page 105)

1. What is the difference between a reflex action and walking?

A **reflex action** is a sudden, automatic response to a stimulus without thinking, controlled mainly by the **spinal cord** (e.g. pulling the hand from a flame). **Walking** is a **voluntary action**: we decide to do it, it is controlled by the **fore-brain** (motor areas of the cerebrum) and its precision and balance by the **cerebellum**; we can start, stop or change it at will. Reflexes are much faster.

2. What happens at the synapse between two neurons?

The electrical impulse reaching the **end of the axon** of the first neuron sets off the release of some **chemicals**. These chemicals **diffuse across the gap (synapse)** and start a **similar electrical impulse in a dendrite of the next neuron**. Thus the electrical signal is converted to a chemical signal and back, and the impulse passes in one direction.

3. Which part of the brain maintains posture and equilibrium of the body?

The **cerebellum**, a part of the hind-brain.

4. How do we detect the smell of an agarbatti (incense stick)?

Molecules of the fragrance spread through the air and reach our nose. The **olfactory receptors** in the nose detect them and produce **nerve impulses**, which travel along the sensory (olfactory) nerve to the **smell area of the fore-brain**. The fore-brain interprets the signal, together with stored memory, and we recognise the smell of the agarbatti.

5. What is the role of the brain in reflex action?

The reflex itself is completed by the **spinal cord** (reflex arc) **without the brain**, so the response is quick. However, the sensory information **also travels to the brain**, which becomes aware of what happened (e.g. we feel the pain afterwards), remembers it and can plan further action (like applying medicine or being careful next time). Some reflexes, like the pupil reflex and sneezing, are controlled by parts of the brain (mid-brain, medulla).

Questions (page 108)

1. What are plant hormones?

Plant hormones (phytohormones) are **chemical substances** produced by plants that coordinate their **growth, development and responses to the environment**. They are made at places away from where they act and **diffuse** to the area of action. Examples: auxin, gibberellins, cytokinins (promote growth), abscisic acid (inhibits growth).

2. How is the movement of leaves of the sensitive plant different from the movement of a shoot towards light?

See the table. The sensitive plant's leaves move in response to **touch**; the movement is **not due to growth**, is very fast and not directional; cells change shape by losing water. The shoot moves towards **light**; the movement is **due to growth** (auxin makes cells on the shady side grow longer), is slow and **directional** (phototropism).

3. Give an example of a plant hormone that promotes growth.

Auxin (helps cells grow longer). Gibberellins (stem growth) and cytokinins (cell division) also promote growth.

4. How do auxins promote the growth of a tendril around a support?

When a tendril touches a support, **auxin moves to the side away from the support**. Cells on that side grow longer (faster) than cells on the side in contact with the support. Because the outer side grows more, the tendril **bends and coils around the support**.

5. Design an experiment to demonstrate hydrotropism.

Fill a glass trough with moist sand/soil and sow a few germinating seeds (gram) in the middle. Bury a **porous clay pot filled with water** on one side of the trough so that only that side is kept moist (do not water the rest). Keep the set-up in uniform conditions for about a week. Carefully remove the sand: the roots are seen to have **bent towards the water pot**. This shows roots are **positively hydrotropic**. (A control trough with no water pot shows roots growing straight down.)

Questions (page 111)

1. How does chemical coordination take place in animals?

Chemical coordination in animals is done by **hormones** secreted by **endocrine (ductless) glands**. The hormones are released **directly into the blood** and carried to **target organs or tissues**, where they produce specific effects (e.g. adrenaline makes the heart beat faster). The timing and amount of hormone released are controlled by **feedback mechanisms**.

2. Why is the use of iodised salt advisable?

Iodine is necessary for the thyroid gland to make **thyroxin**, which regulates carbohydrate, protein and fat metabolism for balanced growth. If our diet lacks iodine, we may suffer from **goitre** (swollen neck). Iodised salt is an easy way to get enough iodine every day.

3. How does our body respond when adrenaline is secreted into the blood?

(1) The **heart beats faster**, supplying more oxygen to the muscles. (2) Muscles around small arteries of the **digestive system and skin contract**, reducing their blood supply and **diverting blood to skeletal muscles**. (3) The **breathing rate increases** because of contractions of the diaphragm and rib muscles. Together these prepare the body to deal with an emergency (fight or flight).

4. Why are some patients of diabetes treated by giving injections of insulin?

In these patients the **pancreas does not secrete enough insulin**, so the **blood sugar level rises**, causing many harmful effects. Insulin is the hormone that **regulates (lowers) blood sugar**. Injecting insulin supplies the missing hormone and brings the sugar level back to normal. (It is injected because, being a protein, it would be digested if taken by mouth.)

In-text questions and activities

i Fig. 6.1 (a): identify the parts of a neuron (i) where information is acquired (ii) through which it travels as an electrical impulse (iii) where it is converted into a chemical signal.

(i) **Dendrite** (dendritic tip). (ii) **Axon** (from dendrite through cell body along the axon). (iii) **Nerve ending** at the end of the axon (axon terminal), at the synapse.

ii Activity 6.1: Is there a difference in how sugar and food taste if your nose is blocked? Why? Do you come across a similar situation when you have a cold?

Yes. Sugar still tastes sweet, but food tastes **flat** and we cannot fully enjoy its flavour. The tongue (gustatory receptors) detects only basic tastes; the rich **flavour** depends on **smell**, detected by olfactory receptors in the nose. With the nose blocked, smell signals do not reach the brain. During a **cold**, mucus blocks the nose, so food seems tasteless in the same way.

– **iii Can you trace the sequence of events which occur when a bright light is focused on your eyes?**

Bright light → **receptors (photoreceptors) in the retina** → **sensory (optic) nerve** → brain (reflex centre) → **motor nerve** → **iris muscles** (effector) → the **pupil constricts** (becomes smaller), letting in less light. It is a reflex action that protects the eye.

– **iv Study the labelled diagram of the human brain. Can we find out the function of each part?**

See the brain table: cerebrum (thinking, senses, voluntary actions), hypothalamus (hunger centre, releasing hormones), pituitary (growth hormone, controls other glands), mid-brain (involuntary actions, reflexes of eyes and head), pons (breathing regulation), medulla (blood pressure, salivation, vomiting, heartbeat), cerebellum (precision, posture, balance), spinal cord (reflexes, conducting messages).

– **v Imagine what would happen if each of these events (walking straight, riding a bicycle, picking up a pencil) failed to take place if we were not thinking about it.**

We would have to **think consciously about every small muscle movement** and every adjustment of balance. We would be slow and clumsy, keep falling over, and could not do two things at once (like walking and talking). The cerebellum frees the thinking brain by handling precision and balance automatically.

– **vi Muscle tissue: based on what we have discussed, what do you think the differences between voluntary and involuntary muscles would be?**

Voluntary muscles get impulses from the **fore-brain** when we decide to move and can be controlled at will (e.g. leg muscles); they are striated and contract quickly. Involuntary muscles (gut, blood vessels, iris; heart) are controlled by the **mid-brain and hind-brain** (and hormones), work on their own without our thinking, and contract slowly and rhythmically.

– **vii Activity 6.2: Have the old parts of the shoot and root changed direction? Are there differences in the direction of the new growth? What can we conclude?**

The **old parts do not change direction**, because they have stopped growing. The **new growth** of the shoot bends **towards light** again and the new root growth bends **away from light**. Conclusion: the shoot is positively phototropic and the root negatively phototropic, and these movements happen by **growth**.

– **viii What would 'hydrotropism' and 'chemotropism' mean? Give examples.**

Hydrotropism: directional growth in response to **water**, e.g. roots growing towards moist soil or a leaking pipe. **Chemotropism**: directional growth in response to **chemicals**, e.g. the **pollen tube growing towards the ovule**.

– **ix Activity 6.3: Identify the endocrine glands in Fig. 6.7.**

Hypothalamus, pineal gland, pituitary gland (in the brain); thyroid and parathyroid glands (neck); thymus (chest); pancreas and adrenal glands (abdomen); testes (male) and ovaries (female). See the gland finder and the list below Table 6.1 for their hormones.

– **x Can you correlate the swollen neck in goitre with the position of the thyroid gland in Fig. 6.7?**

Yes. The thyroid gland lies in the **neck**, in front of the windpipe. When iodine is lacking, the thyroid cannot make enough thyroxin and **enlarges** trying to make more, so the neck looks swollen.

– **xi Activity 6.4: Complete Table 6.1.**

Thyroxin (thyroid); insulin from the pancreas; testosterone: development of male sex organs and features; oestrogen (ovaries); adrenaline: prepares body for emergencies; releasing hormones from the hypothalamus. See the completed table.

Exercises (page 112)

– **1. Which of the following is a plant hormone? (a) Insulin (b) Thyroxin (c) Oestrogen (d) Cytokinin**

(d) Cytokinin.

– **2. The gap between two neurons is called a (a) dendrite (b) synapse (c) axon (d) impulse**

(b) Synapse.

– **3. The brain is responsible for (a) thinking (b) regulating the heart beat (c) balancing the body (d) all of the above**

(d) All of the above.

4. What is the function of receptors in our body? Think of situations where receptors do not work properly. What problems are likely to arise?

Function: receptors are specialised tips of nerve cells, mostly in sense organs, that **detect stimuli** from the environment (light, sound, smell, taste, touch, heat) and start nerve impulses that carry this information to the CNS. **If they do not work properly:** the information will not reach the brain or spinal cord, so we cannot respond. E.g. if heat/pain receptors in the skin fail, we may not pull our hand from a flame and get **burnt**; with a cold (olfactory receptors blocked) we cannot smell or enjoy food, or smell a gas leak; damaged photoreceptors cause poor vision. Our protection and our responses to the environment both suffer.

5. Draw the structure of a neuron and explain its function.

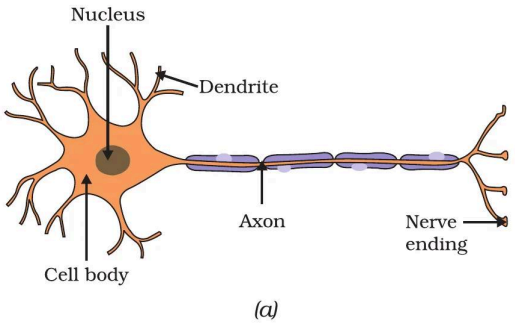


Fig. 6.1 (a) Structure of neuron

Parts: dendrites, cell body with nucleus, axon, nerve ending. **Function:** the neuron is the structural and functional unit of the nervous system; it **conducts information as electrical impulses**. Dendrites acquire information and start an impulse; it travels through the cell body and along the axon; at the nerve ending chemicals are released that cross the synapse to the next neuron, or to a muscle or gland.

6. How does phototropism occur in plants?

Phototropism is the growth movement of plant parts in response to **light**. When light falls on a shoot from one side, **auxin**, made at the shoot tip, **diffuses towards the shady side**. The higher auxin concentration makes cells on the shady side **grow longer** than those on the lit side, so the shoot **bends towards light** (positive phototropism). Roots bend away from light (negative phototropism).

7. Which signals will get disrupted in case of a spinal cord injury?

(1) **Reflex actions** below the injury will not take place, since reflex arcs are in the spinal cord. (2) **Sensory signals** from body parts below the injury will not reach the brain, so there will be no sensation there. (3) **Motor signals** (voluntary commands) from the brain will not reach the muscles below the injury, so those parts cannot be moved (paralysis). In short, both incoming and outgoing messages between the brain and the body below the injury are disrupted.

8. How does chemical coordination occur in plants?

Plants have no nervous system, so they coordinate using **plant hormones**. These are made in one part (e.g. shoot tip) and **diffuse** to where they act. **Auxin** makes cells grow longer and causes bending towards light; **gibberellins** help stem growth; **cytokinins** promote cell division (fruits, seeds); **abscisic acid** inhibits growth and causes wilting. By changing their amounts, the plant controls its growth and responses to light, gravity, touch and water.

9. What is the need for a system of control and coordination in an organism?

(1) Organisms must **detect changes** in the environment and respond with **only the correct movement** (e.g. pull back from a flame, whisper rather than shout). (2) In multicellular organisms, many organs and tissues must **work together** in a proper sequence (e.g. eating involves many muscles and glands). (3) Body conditions such as blood sugar must be **kept balanced**. (4) Growth must happen in the right places. Without control and coordination the body's activities would be haphazard and the organism could not survive.

10. How are involuntary actions and reflex actions different from each other?

See the table. **Involuntary actions** (heartbeat, breathing, salivation, blood pressure) go on without our thinking as part of normal body functions and are controlled by the **mid-brain and hind-brain** (medulla); they need not be triggered by a sudden stimulus. **Reflex actions** are **sudden, quick responses to a specific stimulus**, usually protective (withdrawing the hand from a hot object), and are mostly controlled by the **spinal cord** through a reflex arc.

11. Compare and contrast nervous and hormonal mechanisms for control and coordination in animals.

Compare (similarities): both coordinate body activities and respond to changes; both use chemicals at some stage (synapse chemicals, hormones). **Contrast:** see the nervous vs hormonal table: nervous messages are electrical impulses along nerves, very fast, short-lived, reaching only cells connected by nerves; hormonal messages are chemicals carried in blood, slower, long-lasting, reaching all cells and acting on target organs, and can be released steadily.

12. What is the difference between the manner in which movement takes place in a sensitive plant and the movement in our legs?

See the table. In the **sensitive plant**, touch is communicated from cell to cell by electrical-chemical means (no nerves), and the movement happens because cells **change their water content** and so their shape; no muscles, involuntary. In our **legs**, impulses travel along **nerves** from the brain (voluntary) and **muscle cells** shorten because their **special proteins change shape and arrangement**, pulling the bones.

Diagrams from the NCERT Science textbook for Class 10 (reprint 2026-27). Board questions from CBSE Class 10 Science papers 2023 to 2026.